

```

-> DECLARE counter INT DEFAULT 1;
->
-> WHILE counter <= 5 DO
->     INSERT INTO Employee (EmpName, DeptID)
->     VALUES (CONCAT('Employee', counter), 2);
->     SET counter = counter + 1;
-> END WHILE;
-> END //

```

ERROR 1304 (42000): PROCEDURE InsertEmployeesWhile already exists

```

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL InsertEmployeesWhile();
Query OK, 1 row affected (0.00 sec)

```

```
mysql> SELECT * FROM Employee;
```

EmpID	EmpName	DeptID
1	Employee1	1
2	Employee2	1
3	Employee3	1
4	Employee4	1
5	Employee5	1
6	Employee1	2
7	Employee2	2
8	Employee3	2
9	Employee4	2
10	Employee5	2

10 rows in set (0.00 sec)

```
mysql> |
```

```

-> FROM table_name
-> WHERE condition;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right
syntax to use near 'condition' at line 4
mysql> CREATE VIEW view_name AS
-> SELECT column1, column2
-> FROM table_name
-> WHERE condition;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right
syntax to use near 'condition' at line 4
mysql> CREATE VIEW EmpDept AS
-> SELECT EmpName, DeptName
-> FROM Employee
-> INNER JOIN Department ON Employee.DeptID = Department.DeptID;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM EmpDept;
+-----+-----+
| EmpName | DeptName |
+-----+-----+
| Alice   | Computer Science |
| Charlie | Computer Science |
| Bob     | Mechanical       |
| Eve     | Mechanical       |
| Diana   | Electronics      |
+-----+-----+
5 rows in set (0.00 sec)

mysql> CREATE INDEX idx_deptid ON Employee(DeptID);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> |

```

1	2
2	2
3	1

rows in set (0.00 sec)

```
mysql> SELECT DeptID, COUNT(*) AS EmployeeCount
-> FROM Employee
-> GROUP BY DeptID
-> HAVING COUNT(*) > 1;
```

DeptID	EmployeeCount
1	2
2	2

rows in set (0.00 sec)

```
mysql> SELECT EmpName, Salary
-> FROM Employee
-> ORDER BY Salary DESC;
```

EmpName	Salary
Eve	70000.00
Bob	60000.00
Charlie	55000.00
Alice	50000.00
Diana	45000.00

rows in set (0.00 sec)

```
mysql> |
```

```
mysql> SELECT EmpName
-> FROM Employee
-> WHERE DeptID IN (
->     SELECT DeptID
->     FROM Department
->     WHERE DeptName = 'Computer Science'
-> );
```

EmpName
Alice
Charlie

2 rows in set (0.00 sec)

```
mysql> SELECT EmpName
-> FROM Employee e
-> WHERE EXISTS (
->     SELECT 1
->     FROM Department d
->     WHERE e.DeptID = d.DeptID
->     AND d.DeptName = 'Mechanical'
-> );
```

EmpName
Bob
Eve

2 rows in set (0.00 sec)

Query OK, 1 row affected (0.02 sec)

mysql> USE CompanyDB_Filter;

Database changed

mysql>

mysql> CREATE TABLE Employee (

-> EmpID INT PRIMARY KEY,

-> EmpName VARCHAR(50),

-> DeptID INT

->);

Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Employee (EmpID, EmpName, DeptID) VALUES

-> (101, 'Alice', 1),

-> (102, 'Bob', 2),

-> (103, 'Aman', 1),

-> (104, 'Diana', 3),

-> (105, 'Aaron', 2),

-> (106, 'Anita', 1),

-> (107, 'Brian', 3);

Query OK, 7 rows affected (0.00 sec)

Records: 7 Duplicates: 0 Warnings: 0

mysql> SELECT EmpName FROM Employee WHERE DeptID = 1;

EmpName
Alice
Aman
Anita

3 rows in set (0.00 sec)

```
mysql> DELIMITER ;
mysql> CALL ShowEmployeeNames();
```

```
+-----+
| EmployeeName |
+-----+
| Alice        |
+-----+
1 row in set (0.00 sec)
```

```
+-----+
| EmployeeName |
+-----+
| Bob          |
+-----+
1 row in set (0.01 sec)
```

```
+-----+
| EmployeeName |
+-----+
| Charlie      |
+-----+
1 row in set (0.01 sec)
```

```
+-----+
| EmployeeName |
+-----+
| Diana        |
+-----+
1 row in set (0.02 sec)
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> |
```

Field	Type	Null	Key	Default	Extra
REG	int	NO	PRI	NULL	
NAME	varchar(20)	NO		NULL	
MARKS	int	NO		NULL	

3 rows in set (0.01 sec)

```

MySQL 8.0 Command Line Cli
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 17
Server version: 8.0.43 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE College;
ERROR 1007 (HY000): Can't create database 'college'; database exists
mysql> USE College
Database changed
mysql> CREATE TABLE STUDENT
-> (
->   RegNo INT(3) PRIMARY KEY,
->   Name VARCHAR(15),
->   Gender CHAR(1),
->   DOB DATE,
->   MobileNo BIGINT(10)
-> )
-> ;
ERROR 1050 (42S01): Table 'student' already exists
mysql> DESC STUDENT
-> ;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | NO   | PRI | NULL    |       |
| Name  | varchar(15) | YES |     | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql>

```



```
mysql> SELECT * FROM STUDINFO;
```

REG	NAME	MARKS
1	ANJI	91
2	SURI	93
3	SANJAY	92
21	ANJI	92

```
4 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM STUDINFO WHERE MARKS>91;
```

REG	NAME	MARKS
2	SURI	93
3	SANJAY	92
21	ANJI	92

```
3 rows in set (0.01 sec)
```

```
mysql> |
```

```

mysql>
mysql> CREATE TABLE Department (
->     DeptID INT PRIMARY KEY,
->     DeptName VARCHAR(50)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT SELECT, INSERT ON CompanyDB_Permissions.Department TO 'user1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO Department (DeptID, DeptName) VALUES (1, 'Computer Science');
Query OK, 1 row affected (0.00 sec)

mysql> REVOKE INSERT ON CompanyDB_Permissions.Department FROM 'user1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> -- As user1:
mysql> INSERT INTO Department (DeptID, DeptName) VALUES (2, 'Mechanical');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM Department;
+-----+-----+
| DeptID | DeptName          |
+-----+-----+
|      1 | Computer Science  |
|      2 | Mechanical        |
+-----+-----+
2 rows in set (0.00 sec)

```

```
mysql> DESC STUDINFO  
-> ;
```

Field	Type	Null	Key	Default	Extra
REG	int	NO	PRI	NULL	
NAME	varchar(20)	NO		NULL	
MARKS	int	NO		NULL	

```
3 rows in set (0.01 sec)
```

```
mysql> INSERT INTO STUDINFO ( REG, NAME, MARKS)  
-> VALUES (21,'ANJI',92);  
Query OK, 1 row affected (0.03 sec)
```

```
mysql> SELECT * FROM STUDINFO;
```

REG	NAME	MARKS
21	ANJI	92

```
1 row in set (0.00 sec)
```

```
mysql> |
```

```
    (103, Eve, 2),
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT ProductName, Price FROM Product WHERE Price BETWEEN 30 AND 150;
```

```
+-----+-----+
| ProductName | Price |
+-----+-----+
| Keyboard    | 45.00 |
| Monitor     | 150.00 |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT EmpName FROM Employee WHERE DeptID IN (1, 2);
```

```
+-----+
| EmpName |
+-----+
| Alice   |
| Bob     |
| Diana   |
| Eve     |
+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT AVG(Price) AS AveragePrice FROM Product;
```

```
+-----+
| AveragePrice |
+-----+
| 286.000000   |
+-----+
1 row in set (0.00 sec)
```

```
mysql>
```

```
mysql>
mysql> ROLLBACK TO sp1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM Department;
+-----+-----+
| DeptID | DeptName          |
+-----+-----+
|      1 | Computer Science |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> INSERT INTO Department (DeptID, DeptName)
-> VALUES (4, 'Civil');
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> SELECT * FROM Department;
+-----+-----+
| DeptID | DeptName          |
+-----+-----+
|      1 | Computer Science |
+-----+-----+
1 row in set (0.00 sec)
```

```
Charlie | Computer Science |
-----+-----+
3 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT Employee.EmpName, Department.DeptName
-> FROM Employee
-> LEFT OUTER JOIN Department ON Employee.DeptID = Department.DeptID;
```

```
EmpName | DeptName |
-----+-----+
Alice   | Computer Science |
Bob     | Mechanical       |
Charlie | Computer Science |
Diana   | NULL             |
Eve     | NULL             |
-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT Employee.EmpName, Department.DeptName
-> FROM Employee
-> RIGHT OUTER JOIN Department ON Employee.DeptID = Department.DeptID;
```

```
EmpName | DeptName |
-----+-----+
Charlie | Computer Science |
Alice   | Computer Science |
Bob     | Mechanical       |
NULL    | Electronics      |
-----+-----+
4 rows in set (0.00 sec)
```

```

-> FROM table_name
-> WHERE condition;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right
syntax to use near 'condition' at line 4
mysql> CREATE VIEW view_name AS
-> SELECT column1, column2
-> FROM table_name
-> WHERE condition;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right
syntax to use near 'condition' at line 4
mysql> CREATE VIEW EmpDept AS
-> SELECT EmpName, DeptName
-> FROM Employee
-> INNER JOIN Department ON Employee.DeptID = Department.DeptID;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM EmpDept;
+-----+-----+
| EmpName | DeptName |
+-----+-----+
| Alice   | Computer Science |
| Charlie | Computer Science |
| Bob     | Mechanical       |
| Eve     | Mechanical       |
| Diana   | Electronics      |
+-----+-----+
5 rows in set (0.00 sec)

mysql> CREATE INDEX idx_deptid ON Employee(DeptID);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> |

```

```

->          VALUES (CONCAT('Employee', counter), 3, 4000);
->
->          SET counter = counter + 1;
->
->          IF counter > 5 THEN
->              LEAVE my_loop;
->          END IF;
->      END LOOP;
-> END //

```

Query OK, 0 rows affected (0.02 sec)

```

mysql>
mysql> DELIMITER ;
mysql> CALL InsertLoopEmployees();
Query OK, 1 row affected (0.00 sec)

```

```
mysql> SELECT EmpID, EmpName, DeptID, Salary FROM Employee;
```

EmpID	EmpName	DeptID	Salary
1	Alice	1	6000.00
2	Bob	2	4500.00
3	Charlie	1	2800.00
4	Diana	2	3200.00
5	Employee1	3	4000.00
6	Employee2	3	4000.00
7	Employee3	3	4000.00
8	Employee4	3	4000.00
9	Employee5	3	4000.00

9 rows in set (0.00 sec)

```
mysql>
```



```
mysql> CREATE DATABASE CompanyDB_AutoInc;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> USE CompanyDB_AutoInc;  
Database changed
```

```
mysql> CREATE TABLE Employee (  
->     EmpID INT AUTO_INCREMENT,  
->     EmpName VARCHAR(50),  
->     DeptID INT,  
->     PRIMARY KEY (EmpID)  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Employee (EmpName, DeptID) VALUES  
-> ('Alice', 1),  
-> ('Bob', 2),  
-> ('Charlie', 1);  
Query OK, 3 rows affected (0.00 sec)  
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Employee;
```

EmpID	EmpName	DeptID
1	Alice	1
2	Bob	2
3	Charlie	1

```
3 rows in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO Employee (EmpName, DeptID, Salary) VALUES  
-> ('Alice', 1, 6000),  
-> ('Bob', 2, 4500),  
-> ('Charlie', 1, 3000);
```

```
Query OK, 3 rows affected (0.00 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> DELIMITER //  
mysql>  
mysql> CREATE FUNCTION CalculateSalary(emp_id INT)  
-> RETURNS INT  
-> DETERMINISTIC  
-> BEGIN  
-> DECLARE emp_salary INT;  
-> SELECT Salary INTO emp_salary FROM Employee WHERE EmpID = emp_id;  
-> RETURN emp_salary;  
-> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>  
mysql> DELIMITER ;  
mysql> SELECT CalculateSalary(1);
```

```
+-----+  
| CalculateSalary(1) |  
+-----+  
| 6000 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> |
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE InsertEmployee (
  ->     IN emp_name VARCHAR(50),
  ->     IN dept_id INT
  -> )
  -> BEGIN
  ->     INSERT INTO Employee (EmpName, DeptID)
  ->     VALUES (emp_name, dept_id);
  -> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql> CALL InsertEmployee('John Doe', 1);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> CALL InsertEmployee('Jane Smith', 2);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT * FROM Employee;
+-----+-----+-----+
| EmpID | EmpName   | DeptID |
+-----+-----+-----+
|      1 | John Doe  |      1 |
|      2 | Jane Smith |      2 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```

-> DECLARE counter INT DEFAULT 1;
->
-> WHILE counter <= 5 DO
->     INSERT INTO Employee (EmpName, DeptID)
->     VALUES (CONCAT('Employee', counter), 2);
->     SET counter = counter + 1;
-> END WHILE;
-> END //

```

ERROR 1304 (42000): PROCEDURE InsertEmployeesWhile already exists

```

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL InsertEmployeesWhile();
Query OK, 1 row affected (0.00 sec)

```

```
mysql> SELECT * FROM Employee;
```

EmpID	EmpName	DeptID
1	Employee1	1
2	Employee2	1
3	Employee3	1
4	Employee4	1
5	Employee5	1
6	Employee1	2
7	Employee2	2
8	Employee3	2
9	Employee4	2
10	Employee5	2

10 rows in set (0.00 sec)

```
mysql> |
```

```
mysql> DELIMITER ;
mysql> CALL ShowEmployeeNames();
```

```
+-----+
| EmployeeName |
+-----+
| Alice        |
+-----+
1 row in set (0.00 sec)
```

```
+-----+
| EmployeeName |
+-----+
| Bob          |
+-----+
1 row in set (0.01 sec)
```

```
+-----+
| EmployeeName |
+-----+
| Charlie      |
+-----+
1 row in set (0.01 sec)
```

```
+-----+
| EmployeeName |
+-----+
| Diana        |
+-----+
1 row in set (0.02 sec)
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> |
```

```

->          VALUES (CONCAT('Employee', counter), 3, 4000);
->
->          SET counter = counter + 1;
->
->          IF counter > 5 THEN
->              LEAVE my_loop;
->          END IF;
->      END LOOP;
-> END //

```

Query OK, 0 rows affected (0.02 sec)

```

mysql>
mysql> DELIMITER ;
mysql> CALL InsertLoopEmployees();
Query OK, 1 row affected (0.00 sec)

```

```

mysql> SELECT EmpID, EmpName, DeptID, Salary FROM Employee;

```

EmpID	EmpName	DeptID	Salary
1	Alice	1	6000.00
2	Bob	2	4500.00
3	Charlie	1	2800.00
4	Diana	2	3200.00
5	Employee1	3	4000.00
6	Employee2	3	4000.00
7	Employee3	3	4000.00
8	Employee4	3	4000.00
9	Employee5	3	4000.00

9 rows in set (0.00 sec)

```
mysql>
```

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE InsertEmployee (  
->     IN emp_name VARCHAR(50),  
->     IN dept_id INT  
-> )  
-> BEGIN  
->     INSERT INTO Employee (EmpName, DeptID)  
->     VALUES (emp_name, dept_id);  
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql> CALL InsertEmployee('John Doe', 1);
```

Query OK, 1 row affected (0.00 sec)

```
mysql> CALL InsertEmployee('Jane Smith', 2);
```

Query OK, 1 row affected (0.00 sec)

```
mysql> SELECT * FROM Employee;
```

EmpID	EmpName	DeptID
1	John Doe	1
2	Jane Smith	2

2 rows in set (0.00 sec)

```
| Charlie | Computer Science |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT Employee.EmpName, Department.DeptName
-> FROM Employee
-> LEFT OUTER JOIN Department ON Employee.DeptID = Department.DeptID;
```

```
+-----+-----+
| EmpName | DeptName |
+-----+-----+
| Alice   | Computer Science |
| Bob     | Mechanical       |
| Charlie | Computer Science |
| Diana   | NULL             |
| Eve     | NULL             |
+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT Employee.EmpName, Department.DeptName
-> FROM Employee
-> RIGHT OUTER JOIN Department ON Employee.DeptID = Department.DeptID;
```

```
+-----+-----+
| EmpName | DeptName |
+-----+-----+
| Charlie | Computer Science |
| Alice   | Computer Science |
| Bob     | Mechanical       |
| NULL    | Electronics      |
+-----+-----+
4 rows in set (0.00 sec)
```



```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO Employee (EmpName, DeptID, Salary) VALUES
-> ('Alice', 1, 6000),
-> ('Bob', 2, 4500),
-> ('Charlie', 1, 3000);
```

```
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION CalculateSalary(emp_id INT)
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE emp_salary INT;
->     SELECT Salary INTO emp_salary FROM Employee WHERE EmpID = emp_id;
->     RETURN emp_salary;
-> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql> SELECT CalculateSalary(1);
```

```
+-----+
| CalculateSalary(1) |
+-----+
|                6000 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> |
```

```
mysql> SELECT * FROM STUDINFO;
```

REG	NAME	MARKS
1	ANJI	91
2	SURI	93
3	SANJAY	92
21	ANJI	92

4 rows in set (0.01 sec)

```
mysql> SELECT * FROM STUDINFO WHERE MARKS>91;
```

REG	NAME	MARKS
2	SURI	93
3	SANJAY	92
21	ANJI	92

3 rows in set (0.01 sec)

```
mysql> |
```

```
mysql> DESC STUDINFO
-> ;
```

Field	Type	Null	Key	Default	Extra
REG	int	NO	PRI	NULL	
NAME	varchar(20)	NO		NULL	
MARKS	int	NO		NULL	

```
3 rows in set (0.01 sec)
```

```
mysql>
mysql> ROLLBACK TO sp1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM Department;
+-----+-----+
| DeptID | DeptName          |
+-----+-----+
|      1 | Computer Science |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> INSERT INTO Department (DeptID, DeptName)
      -> VALUES (4, 'Civil');
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> SELECT * FROM Department;
+-----+-----+
| DeptID | DeptName          |
+-----+-----+
|      1 | Computer Science |
+-----+-----+
1 row in set (0.00 sec)
```

```

mysql>
mysql> CREATE TABLE Department (
->     DeptID INT PRIMARY KEY,
->     DeptName VARCHAR(50)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT SELECT, INSERT ON CompanyDB_Permissions.Department TO 'user1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO Department (DeptID, DeptName) VALUES (1, 'Computer Science');
Query OK, 1 row affected (0.00 sec)

mysql> REVOKE INSERT ON CompanyDB_Permissions.Department FROM 'user1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> -- As user1:
mysql> INSERT INTO Department (DeptID, DeptName) VALUES (2, 'Mechanical');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM Department;
+-----+-----+
| DeptID | DeptName          |
+-----+-----+
|      1 | Computer Science  |
|      2 | Mechanical        |
+-----+-----+
2 rows in set (0.00 sec)

```

```

MySQL 8.0 Command Line Cli  X  +  v

Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 17
Server version: 8.0.43 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE College;
ERROR 1007 (HY000): Can't create database 'college'; database exists
mysql> USE College
Database changed
mysql> CREATE TABLE STUDENT
-> (
-> RegNo INT(3)PRIMARY KEY,
-> Name VARCHAR(15),
-> Gender CHAR(1),
-> DOB DATE,
-> MobileNo BIGINT(10)
-> )
-> ;
ERROR 1050 (42S01): Table 'student' already exists
mysql> DESC STUDENT
-> ;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | NO   | PRI | NULL    |       |
| Name  | varchar(15) | YES |     | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql>

```

```
mysql> DESC STUDINFO
-> ;
```

Field	Type	Null	Key	Default	Extra
REG	int	NO	PRI	NULL	
NAME	varchar(20)	NO		NULL	
MARKS	int	NO		NULL	

```
3 rows in set (0.01 sec)
```

```
mysql> INSERT INTO STUDINFO ( REG, NAME, MARKS)
-> VALUES (21,'ANJI',92);
```

```
Query OK, 1 row affected (0.03 sec)
```

```
mysql> SELECT * FROM STUDINFO;
```

REG	NAME	MARKS
21	ANJI	92

```
1 row in set (0.00 sec)
```

```
mysql> |
```

```
    (103, Eve, 2),
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT ProductName, Price FROM Product WHERE Price BETWEEN 30 AND 150;
```

```
+-----+-----+
| ProductName | Price |
+-----+-----+
| Keyboard    | 45.00 |
| Monitor     | 150.00 |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT EmpName FROM Employee WHERE DeptID IN (1, 2);
```

```
+-----+
| EmpName |
+-----+
| Alice   |
| Bob     |
| Diana   |
| Eve     |
+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT AVG(Price) AS AveragePrice FROM Product;
```

```
+-----+
| AveragePrice |
+-----+
| 286.000000   |
+-----+
1 row in set (0.00 sec)
```

```
mysql>
```


Query OK, 1 row affected (0.02 sec)

mysql> USE CompanyDB_Filter;
Database changed

mysql>
mysql> CREATE TABLE Employee (
-> EmpID INT PRIMARY KEY,
-> EmpName VARCHAR(50),
-> DeptID INT
->);

Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Employee (EmpID, EmpName, DeptID) VALUES
-> (101, 'Alice', 1),
-> (102, 'Bob', 2),
-> (103, 'Aman', 1),
-> (104, 'Diana', 3),
-> (105, 'Aaron', 2),
-> (106, 'Anita', 1),
-> (107, 'Brian', 3);

Query OK, 7 rows affected (0.00 sec)

Records: 7 Duplicates: 0 Warnings: 0

mysql> SELECT EmpName FROM Employee WHERE DeptID = 1;

EmpName
Alice
Aman
Anita

3 rows in set (0.00 sec)

```
mysql> CREATE DATABASE CompanyDB_AutoInc;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> USE CompanyDB_AutoInc;  
Database changed
```

```
mysql> CREATE TABLE Employee (  
->     EmpID INT AUTO_INCREMENT,  
->     EmpName VARCHAR(50),  
->     DeptID INT,  
->     PRIMARY KEY (EmpID)  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Employee (EmpName, DeptID) VALUES  
-> ('Alice', 1),  
-> ('Bob', 2),  
-> ('Charlie', 1);  
Query OK, 3 rows affected (0.00 sec)  
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Employee;
```

EmpID	EmpName	DeptID
1	Alice	1
2	Bob	2
3	Charlie	1

```
3 rows in set (0.00 sec)
```

1	2
2	2
3	1

rows in set (0.00 sec)

```
mysql> SELECT DeptID, COUNT(*) AS EmployeeCount
-> FROM Employee
-> GROUP BY DeptID
-> HAVING COUNT(*) > 1;
```

DeptID	EmployeeCount
1	2
2	2

rows in set (0.00 sec)

```
mysql> SELECT EmpName, Salary
-> FROM Employee
-> ORDER BY Salary DESC;
```

EmpName	Salary
Eve	70000.00
Bob	60000.00
Charlie	55000.00
Alice	50000.00
Diana	45000.00

rows in set (0.00 sec)

mysql> |

```
mysql> SELECT EmpName
-> FROM Employee
-> WHERE DeptID IN (
->     SELECT DeptID
->     FROM Department
->     WHERE DeptName = 'Computer Science'
-> );
```

EmpName
Alice
Charlie

2 rows in set (0.00 sec)

```
mysql> SELECT EmpName
-> FROM Employee e
-> WHERE EXISTS (
->     SELECT 1
->     FROM Department d
->     WHERE e.DeptID = d.DeptID
->     AND d.DeptName = 'Mechanical'
-> );
```

EmpName
Bob
Eve

2 rows in set (0.00 sec)